

OMAR ELGENDY

Machine Learning Engineer | Data Scientist

+971-558980116 • omaratef3221@gmail.com • linkedin.com/in/omaratef3221 • github.com/omaratef3221 • omarelgendy.dev
• Dubai, United Arab Emirates

Summary

Machine Learning Engineer and AI Researcher with 6+ years building and deploying production ML systems, specialized in LLMs, RAG, and agentic AI. Track record of taking generative-AI applications from prototype to production: fine-tuning transformers (LoRA/QLoRA), building RAG systems and AI agents, and deploying them on end-to-end MLOps pipelines across AWS and GCP, all grounded in a deep classical-ML foundation spanning gradient boosting and ML Algorithms (e.g. XGBoost), ensemble methods, and time-series forecasting on structured data. Published researcher with 6 peer-reviewed papers, 550+ citations, and h-index 4, including a top-0.1%-cited Computer Science paper on healthcare AI.

Experience

Almosafer

Dubai, United Arab Emirates

Machine Learning Specialist

09/2025 - Present

- Built and tested a production voice AI agent (Retell AI) with LLM-driven dialogue management and tool use to assist customer service, optimizing response quality and call-handling latency for real-time customer interactions.
- Built and deployed a production ML forecasting time series system for flight add-on revenue, increasing add-on revenue by 55%.
- Designed and supported a post-call GenAI pipeline using LLM-based summarization, classification, and entity extraction to convert raw call transcripts into structured records automatically.
- Developed and tested an XGBoost financial-forecasting model that cut prediction error (MAPE) by 80% versus the prior baseline, improving budgeting accuracy.
- Architected and validated end-to-end MLOps pipelines on AWS SageMaker and MLflow, automating model training, versioning, and deployment at scale.

University of Sharjah

Sharjah, United Arab Emirates

Research Assistant (Part-time)

06/2024 - 01/2026

- Built the AI chatbot ("Welly") for My FarmWell – a University of Sharjah × IBM agricultural app (IBM Sustainability Accelerator) delivering a bilingual Arabic/English RAG assistant that answers farmer questions on irrigation, water conservation, and local regulations.
- Engineered and launched the retrieval-augmented generation pipeline over region-specific water, climate, and groundwater data, grounding LLM responses in IBM Environmental Intelligence and IBM Cloud sources for accurate, farm-level guidance.
- Additional work: collaborated on writing 2 papers under review on ML for groundwater-level forecasting (time-series, AutoML), 2026.

Virtue Therapy

Dubai, United Arab Emirates

Machine Learning Engineer

03/2025 - 03/2025

- Developed and integrated production-grade LLM applications for chatbots, document Q&A, and intelligent assistants using Transformers, LangChain, and RAG, achieving over 85% accuracy in real-world tasks.
- Deployed serverless AI APIs using Docker and AWS Lambda, reducing hosting costs by 25% and maintaining sub-3-second response latency under 1000 RPS.
- Built scalable MLOps pipelines on AWS SageMaker, MLflow, Airflow, and TorchServe, cutting deployment time by 30% and automating versioning, CI/CD, and monitoring.
- Fine-tuned transformer models with QLoRA and LoRA across tourism, agriculture, and healthcare, improving task accuracy by up to 18%.
- Engineered and explored optimized prompts and RLHF to enhance LLM summarization and multi-turn dialogue performance.

Virtue Therapy

Dubai, United Arab Emirates

Junior Machine Learning Engineer

02/2021 - 01/2023

- Built and deployed machine learning models for user behavior prediction using mood analysis, psychological profiling, and sentiment analysis, improving engagement insights by 30%.
- Developed and trained classic machine learning models (random forests, XGBoost, SVMs) for structured data tasks, improving prediction accuracy by up to 22% over baseline.
- Led end-to-end ML workflows – data preprocessing, EDA, model training, tuning, and evaluation – using Scikit-learn, TensorFlow, and PyTorch, increasing model precision by 20%.
- Implemented and orchestrated experiment tracking and model versioning for better reproducibility and lifecycle management.

Experience

University of Sharjah	Sharjah, United Arab Emirates
Research Assistant	08/2020 - 02/2021
- Participated and authored 6+ peer-reviewed AI research papers—including 2 in Q1-ranked journals—with 490+ total citations and a h-index of 4, focusing on deep learning, NLP, and clinical applications. - Developed an optimized Arabic text summarization approach in Master's thesis by fine-tuning transformer models with Radial Basis Function Neural Networks, reducing training time by 45% while maintaining high performance. - Developed and contributed BERT-based models for Arabic fake news detection, achieving a 90%+ classification accuracy and published in high-impact NLP journals. - Planned and evaluated Deep Learning architectures for Alzheimer's detection using MRI and for heart failure prediction using structured medical datasets.	

Skills

ML & AI: Python · TensorFlow · PyTorch · XGBoost · Scikit-learn · Keras · Neural Networks · C/C++
LLMs & NLP: Transformers · LangChain · LangGraph · RAG · ReAct · NLTK · Fine-Tuning · LoRA · QLoRA · RLHF
MLOps & Cloud: AWS · AWS SageMaker · MLflow · Airflow · Docker · CI/CD · GCP · Azure · TorchServe · Databricks · Kubernetes
Monitoring & Infra: Prometheus · Grafana · Bash · Linux · Git · GitHub · Terraform
Data & Backend: SQL · PostgreSQL · MongoDB · FastAPI · Flask · NumPy · Pandas
Soft skills: Decision-Making · Teamwork · Creativity · Adaptability · Continuous Learning · Leadership · Problem-Solving · Emotional Intelligence · Ownership · Agile · Attention to Detail

Education

University of Sharjah	Sharjah, United Arab Emirates
M.Sc. in Computer Engineering, minor Artificial Intelligence	09/2020 - 12/2024
Thesis: Optimizing Large Language Models for Arabic Text Summarization utilizing RBF Networks	
University of Sharjah	Sharjah, United Arab Emirates
B.Sc. in Computer Engineering	09/2016 - 08/2020
Senior Project: EEG-based Computer Command Input for paralyzed people	

Languages

Arabic Native	English Fluent
---------------	----------------

Certifications

AWS Certified Machine Learning Specialty — Amazon Web Services (AWS)
AWS Certified Cloud Practitioner — Amazon Web Services (AWS)

Publications

The 2024 OkIP International Conference on Advances in Health Information Technology (AHIT) - OkIP Books	2024
Text Toxicity Level Detection using Deep Contextualized Embedding Models <i>O. Elgendy, A. B. Nassif, and B. Soudan</i>	
2025 18th International Conference on Development in eSystem Engineering (DeSE)	2025
Enhancing RLHF in LLMs: Comparing BERT, XGBoost, and Deep Models for Bias Detection <i>W. Torfa, O. Elgendy, and A. B. Nassif</i>	

Publications

The 2024 OkIP International Conference on Advances in Health Information Technology (AHIT) - OkIP Books Heart Failure Prediction using Machine learning with Meta-heuristic feature selection techniques <i>O. Elgendy, A. B. Nassif, and B. Soudan</i>	2024
2023 Advances in Science and Engineering Technology International Conferences (ASET) - IEEE Alzheimer Detection using Different Deep Learning Methods with MRI Images <i>O. Elgendy and A. B. Nassif</i>	2023
Neural Computing and Applications - Springer - Q1 Arabic fake news detection based on deep contextualized embedding models <i>A. B. Nassif, A. Elnagar, O. Elgendy, and Y. Afadar</i>	2022
Artificial intelligence in medicine - ScienceDirect - Q1 Breast cancer detection using artificial intelligence techniques: A systematic literature review <i>A. B. Nassif, M. A. Talib, Q. Nasir, Y. Afadar, and O. Elgendy</i>	2022